# Unit 7 Quiz: Probability Distributions KEY

Name _____

Show your work.

## 7.1 - Probability Distributions

1. **I check if a probability distribution is valid, and I draw it as a graph.**

   (a) Is this table a valid probability distribution? Explain.

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.5 | 0.4 | $-0.1$ |

Not valid. The probabilities add to 1, but $-0.1$ is less than 0. A probability cannot be negative.

Let $X$ = the number of times the school bus is late in one week.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.35 | 0.30 | ? | 0.10 | 0.05 |

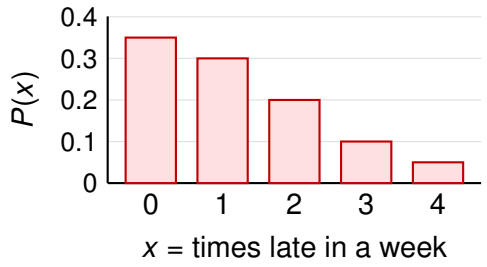
   (b) Find the missing probability, $P(2)$.

$1 - (0.35 + 0.30 + 0.10 + 0.05)$
$= 1 - 0.80 = 0.20$

   (c) Find $P(X \geq 2)$.

$0.20 + 0.10 + 0.05 = 0.35$

   (d) Draw a bar graph of the distribution.




2. **I explain what a random variable is, and I make theoretical and empirical distributions.**

A box has 10 cards. 5 cards show the number 1. 3 cards show 2. 2 cards show 5. You draw one card without looking. Let $X$ = the number on the card.

   (a) Why is $X$ a random variable?

$X$ is a number that comes from a chance process. You do not know which card you will draw.

   (b) Make the **theoretical** probability distribution of $X$.

| $x$ | 1 | 2 | 5 |
|---|---|---|---|
| $P(x)$ | $\frac{5}{10} = 0.5$ | $\frac{3}{10} = 0.3$ | $\frac{2}{10} = 0.2$ |

Ms. Diaz's class drew a card 50 times. They put the card back each time. They got a 1 on 22 draws, a 2 on 18 draws, and a 5 on 10 draws.

   (c) Make the **empirical** probability distribution of $X$.

| $x$ | 1 | 2 | 5 |
|---|---|---|---|
| $P(x)$ | $\frac{22}{50} = 0.44$ | $\frac{18}{50} = 0.36$ | $\frac{10}{50} = 0.20$ |

   (d) Why are the two distributions not the same?

The empirical one comes from only 50 draws. Chance makes the counts vary. With more draws, it gets closer to the theoretical one.

## 7.2 - Expected Value

3. **I find the expected value of a random variable, and I explain what it means.**

A basketball team scores 0, 1, 2, or 3 points each time it has the ball. Let $X$ = the number of points it scores on one turn with the ball.

| $x$ (points) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(x)$ | 0.45 | 0.05 | 0.35 | 0.15 |
| $x \cdot P(x)$ | 0 | 0.05 | 0.70 | 0.45 |

(a) Fill in the last row of the table.

(b) Find $E(X)$.

$E(X) = \Sigma\, x \cdot P(x)$
$= 0 + 0.05 + 0.70 + 0.45 = 1.2$ points

(c) Which statement best describes the meaning of $E(X)$? Circle the letter.

A The team scores exactly 1.2 points each time it has the ball.
(B) Over many turns with the ball, the team scores about 1.2 points per turn, on average.
C The team scores 1 point more often than any other number of points.
D Over many turns with the ball, the team scores 1.2 points in total.

(d) The team has the ball 60 times in a game. About how many points do you expect it to score?

$60 \times 1.2 = 72$ points

4. **I decide if a method of choosing is fair, and I decide if a game is fair.**

Lin, Omar, and Sara need to choose who presents first.

(a) Which method is fair? Circle the letter.

A Roll a number cube. 1 or 2: Lin. 3: Omar. 4, 5, or 6: Sara.
Ⓑ Roll a number cube. 1 or 6: Lin. 2 or 5: Omar. 3 or 4: Sara.
C Flip a coin 3 times. 3 heads: Lin. 3 tails: Omar. Anything else: Sara.
D Spin a spinner with 5 equal parts. 2 parts: Lin. 2 parts: Omar. 1 part: Sara.

(b) Explain your choice.

Each person gets 2 of the 6 numbers, so each has $\frac{2}{6} = \frac{1}{3}$.

At a school fair game booth, you roll a number cube. You win \$4 if you roll a 1 or a 2. You pay \$3 for any other number.

(c) Find the expected value of what you win.

$$E(X) = 4 \cdot \frac{2}{6} + (-3) \cdot \frac{4}{6} = \frac{8}{6} - \frac{12}{6} = -\frac{4}{6} \approx -\$0.67$$

(d) Is the game fair? Explain.

No. The expected value is not 0. On average you lose about 67 cents a game.

(e) How much should you win for a 1 or a 2 to make the game fair?

$w \cdot \frac{2}{6} - \frac{12}{6} = 0$, so $w = 6$.
You should win \$6.

## 7.3 - Expected Payoff

5. **I find the net gain for each outcome, and I find the expected payoff of a lottery, game, or plan.**

   The soccer team sells 100 raffle tickets for $5 each. One ticket wins $200. Five tickets win $20 each. The rest win nothing.

   (a) Fill in the table.

| Outcome | Net gain $x$ | $P(x)$ | $x \cdot P(x)$ |
|---|---|---|---|
| Win $200 | $200 - 5 = 195$ | $\frac{1}{100} = 0.01$ | 1.95 |
| Win $20 | $20 - 5 = 15$ | $\frac{5}{100} = 0.05$ | 0.75 |
| Win nothing | $-5$ | $\frac{94}{100} = 0.94$ | $-4.70$ |

   (b) Find the expected payoff.

   $1.95 + 0.75 - 4.70 = -\$2.00$

   (c) What does the expected payoff mean?

   On average, a buyer loses $2 per ticket.

   (d) The team sells all 100 tickets. How much money does the team raise?

   $100 \times \$2 = \$200$ (or $\$500 - \$200 - \$100 = \$200$)

6. **I use expected value, and risk, to make a decision.**

   A tablet screen costs $180 to fix. The chance that a screen breaks in one year is 0.15. A one-year protection plan costs $35.

   (a) Find the expected cost to fix the screen with no plan.

   $0.15 \cdot 180 = \$27$

   (b) Which is cheaper on average: buying the plan or not? By how much?

   Not buying it. $\$35 - \$27 = \$8$ cheaper on average.

   (c) Give a reason someone might still buy the plan.

   They could not pay a sudden $180 bill. The plan removes that risk.

   (d) The company sells 2,000 of these plans. About how much money does it expect to make?

   Each plan: $\$35 - \$27 = \$8$.
   $2{,}000 \times 8 = \$16{,}000$